

Element L

After completing the design process it is clear that there are many things that can be improved in order to create a better final product and processes that could have aided in the building of the sink. One main improvement that could have been made is evident in the flow of the water. A critical issue we ran into is that the tube that connects the two water jugs is not optimal for the flow rate of the jugs when both turned on. When the sink is turned on the water flows from one end to the other skipping over the hole causing backflow out of the back of the sink and stressing the connection leading to instability and the tube becoming loose and sometimes falling off. Although we found some workarounds such as making the hole bigger and adding more adhesive to keep the tube connected better, the issue is still persistent at a lesser noticeability. If we were to remake this in the future, we would redesign the system with a longer tubing system in order to reduce the stress added on the bending, additionally the longer tubing would negate the backflow issue by having more space for the water in case of excess backflow.

Another issue that was persistent with this design was the comfortability when carrying the sink. We met our goal to have the sink be lightweight enough for the average person to carry without issues. While that was effective, comfortability is another element of carrying a sink that was not really taken into consideration as it was not as important as our other design criteria. In a future design of this project it would be good to incorporate another handle on the sides other than using only the handles built into the plastic bin. This would lead to a more comfortable sink for moving it around, and also it would alleviate stress for the person carrying the sink as it would not take as much effort to move it. This would lead to a more efficient design as the struggle would be relieved, allowing for more portability options.

Sink portability is an additional aspect that brought challenges in our project. In the real world, portable sinks are not very common, mainly because lugging around a piping system, storage area, and disposal system will add lots of extra weight, all while trying to use durable materials. For these reasons, most companies look for other options when considering purchasing a portable sink. Even then, most portable sinks use wheels so that the user does not have to carry the utility. However, our stakeholders requested a smaller portable sink which could be placed on top of a student's desk. Unfortunately, with our goals in mind, there is no easy way around these challenges. We believe we came pretty close to the optimal solution, with a lightweight plastic container as our base. With that being said, other methods should definitely be considered when designing an updated version. A smaller sink is definitely possible. Even with our design, we barely reached our acceptable criteria for weight goals. The wood on top of our base definitely added some extra weight, and it's hard to decide whether or not it was the right decision. We recommend that anyone making another design seriously

considers the weight element, and as they construct their design, look for additional ways to take the total weight down.

Water storage and disposal capacity goals were two targets we set too high. Before designing, we should have asked more specific questions and done more research about how much water we will need for one hand wash. Looking back, we may have prepared for over four times as much water storage and flow capacity as we needed. While it seems like having more than enough water is a good problem to have, this made our sink more unwieldy and heavy. Due to this, we have to instruct the teachers to only fill the water jugs half way up to reduce weight issues. In hindsight, we probably should have purchased a smaller bin for waste. This would not only be closer to our weight goals, but also produce a smaller sink which would be easier to carry. For future projects, we would definitely recommend making additional tests and research to ensure the proper amount of water metrics.

Something our group overlooked was the methods in which we dispose of the water. We figured that the water disposal would be simple, and that we should just focus on the main functionality of the sink. Unfortunately, our method of water disposal requires complete disassembly of the sink before the sink can be emptied. In a new prototype, we would recommend installing a plug or other method to easily dispose of water.